

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and operating large-scale distributed systems. Built infrastructure powering video streaming at **YouTube**, marketplace platforms at **Whatnot**, and search systems at **Google**. Owned end-to-end services including configuration management, on-call rotations, and cross-functional partnerships with security and product teams. Led teams from 2 to 8 engineers, delivered multi-region deployments, and drove reliability improvements across production environments. Next focus on infrastructure automation and platform engineering at scale. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL, Golang, Ruby

Distributed Systems: Large-scale architecture, multi-region deployment, high-throughput systems, fault tolerance, service mesh

Cloud Platforms: AWS, GCP, Azure, multi-cloud infrastructure, infrastructure governance, cost optimization

Infrastructure as Code: Terraform, Pulumi, Ansible, configuration management, automated provisioning

Programming: Go, Python, C#, .NET, Java, PowerShell, scripting and automation

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Architected video infrastructure components serving millions of concurrent users with **Go** and **Python**, achieving sub-second latency targets across global edge networks.
- Designed search ranking infrastructure processing billions of daily queries, implementing caching layers that reduced backend load by 40 percent.
- Established reliability standards including SLO definitions, error budget tracking, and post-incident review processes for video serving pipelines.
- Mentored engineers on distributed systems patterns, code review practices, and operational procedures for 24/7 on-call rotations.
- Led technical design reviews for cross-functional initiatives spanning content delivery, recommendation systems, and platform stability.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built marketplace platform infrastructure handling live auctions and real-time bidding using **Go** and **Kubernetes** on **AWS**.
- Established observability stack with Prometheus, Grafana, and structured logging, enabling faster incident triage and root cause analysis.
- Implemented **Terraform** modules for VPC networking, ECS clusters, and RDS instances across staging and production environments.
- Collaborated with security team on IAM policies, network segmentation, and compliance requirements for payment processing systems.
- Designed multi-region deployment pipeline reducing deployment failures by 35 percent while maintaining blue-green rollout capabilities.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led search infrastructure team serving billions of daily queries, owning services from index generation through query serving with **C++** and **Go**.

- Managed team scaling from 2 to 8 engineers, conducting hiring, performance reviews, and career development across multiple seniority levels.
- Reduced p99 latency by 45 percent through query optimization, caching strategies, and sharding improvements across distributed index nodes.
- Drove migration of on-premises workloads to **GCP**, implementing **Terraform** automation for account lifecycle management.
- Delivered zero-downtime infrastructure upgrades across 12 data centers, coordinating cross-team dependencies and rollback procedures.

Software Engineer | Google | 02/2011 - 02/2015

- Developed backend services for search ranking infrastructure processing millions of requests per second in **C++**.
- Built automated testing frameworks and continuous integration pipelines improving code coverage and reducing integration bugs.
- Participated in on-call rotations, triaging production incidents and implementing permanent fixes for recurring failure modes.
- Optimized memory allocation patterns in serving infrastructure, yielding 20 percent reduction in hardware costs.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and shipped features in **Windows Server 2008 R2** using **C#** and **.NET** framework for enterprise customers.
- Built administrative tools and PowerShell cmdlets for server configuration, reducing manual intervention in data center operations.
- Automated testing and deployment processes with TFS, cutting release cycle time from weeks to days.
- Investigated kernel-level bugs and crashes, working with Windows core engineering teams on patches and hotfixes.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal web applications for employee self-service using **ASP.NET** and **C#**, improving HR workflow efficiency.
- Documented system requirements and participated in code reviews with full-time engineers on enterprise software projects.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007